

AQUAVIEW Indexing Request — HHP Prediction

Datasets to add to the AQUAVIEW MCP catalog to support Hurricane Heat Potential (HHP / TCHP) prediction work.

Author: Suramya Angdembay, Institute for Advanced Analytics and Society, USM. **Date:** 2026-04-29.

Status: all gaps below were verified against the live AQUAVIEW MCP (catalog endpoint <https://mcp.aquaview.org/mcp>) using multi-strategy keyword and collection probes. Items already in AQUAVIEW are not re-listed here. The full corrected inventory of what is already indexed (notably the comprehensive AOML hurricane archive in collection NOAA_AOML_HDB) is available in the project's working notes.

Tier 1 — Real gaps with HHP impact

Highest-priority missing items for the active HHP pipeline. Items rank by how blocked the pipeline is without them.

CMEMS GLORYS-family TEMPERATURE at depth (T(z))	salinity_model_g_<year> exists but no temperature_model_g_<year> partner. Verified — q="temperature_model" returns 0.	Without GLORYS T(z), we cannot compute GLORYS-derived TCHP as an independent comparator to RTOFS-derived TCHP. The salinity-only ingestion is presumably for Argo S calibration; we need the partner T fields.	Marine Copernicus product page (GL OBAL MULTIYEAR_PHY_001_030)	copernicus-marine toolbox docs
CMEMS GLORYS-family CURRENTS (u, v at depth)	Also missing as partner to salinity_model_g_* (same product family ships T, S, u, v together).	Lower priority for HHP scalar output, but needed if we extend to MLD / eddy-correction work or to a Loop-Current intrusion mode.	Marine Copernicus product page	CMEMS PUM (Product User Manual)
Real-time 3D RTOFS (rtofs_glo_3dz_*)	AQUAVIEW UAF aggregator has 2 RTOFS items, both 2D surface forecasts. The 3D f006/f024 NetCDF files we use directly from S3 are not STAC items.	Direct AQUAVIEW discovery of the data we already pull would let any future user replicate the pipeline without bespoke S3 path knowledge.	AWS Open Data Registry — NOAA Global RTOFS	NCO RTOFS product page

Tier 2 — Climate baselines and out-of-basin comparators

Lower urgency for the active demo but high value once the project extends to seasonal validation, multi-storm benchmarking, or cross-model checks.

ECMWF ORAS5	Confirmed missing — q="ORAS5" returns 0.	OHC700 climate-scale comparator. Independent NEMOVAR 3DVar-FGAT reanalysis line; useful once we extend beyond Atlantic-basin focus.	Climate Data Store — ORAS5 catalog	ECMWF — OCEAN5 reanalysis system paper

Met Office EN4	Confirmed missing (Hadley / EN4 / ENACT keyword probes all 0 real hits).	Gridded in-situ-OI T/S, the canonical "Argo climatology" — needed for the seasonal validation demo (winter / summer OHC means).	UK Met Office — EN4 download page	EN4 documentation portal
ISAS (LOPS / Ifremer)	Confirmed missing.	Independent European Argo-OI gridded T/S, ½° monthly with MLD diagnosed. Cross-product validation against EN4 and Roemmich-Gilson.	SEANOE — ISAS dataset DOI	Coriolis — ISAS product page
Roemmich-Gilson Argo Climatology	Confirmed missing across 7 keyword strategies and full APDRC inventory.	Reference Argo-only gridded T/S climatology (1° monthly). Pre-computed gridded fields solve the "interpolate Argo to a uniform grid" step Gregg specifically asked for.	Scripps SIO — RG Argo Climatology	Roemmich & Gilson 2009 — original paper (DOI)
NRL HYCOM 2019-present	AQUAVIEW UAF has 31 HYCOM items but only the 2016-2018 GLBu0.08 expt 91.2 reanalysis.	Operational HYCOM updates regularly (now ESPC-D v02). Adding the recent stream gives an independent ocean model for cross-model validation.	HYCOM data server — ESPC-D v02 global analysis	Barton et al. 2021 — Navy ESPC system paper
JMA TCHP for Western Pacific	Confirmed 0 hits.	Out-of-basin sanity check. The only non-US operational TCHP equivalent currently published, covering the most active typhoon basin. Useful for demonstrating method transfer.	JMA NEAR-GOOS database (RSMC Tokyo)	JMA Tropical Cyclone Heat Potential page

Distribution mechanism note

None of these datasets are natively distributed via ERDDAP. AQUAVIEW already runs an ERDDAP scraper (most of NOAA_AOML_HDB was ingested that way), but the missing items above sit on different protocols:

CMEMS GLORYS / GLO12	copernicus-marine Python client (CMEMS account required)	Lowest-effort: AOML already runs the CMEMS pipeline for salinity_model_g; adding T and u/v variables to the existing fetch is incremental work.
ECMWF ORAS5	cdsapi Python client (CDS account required)	New ingestion path. Monthly cadence, well-documented API.
Met Office EN4	Plain HTTPS / FTP of NetCDF files	Anonymous; simple requests-based fetcher.
ISAS (Ifremer)	SEANOE DOI + Coriolis FTP	Anonymous; per-version DOI updates.

Roemmich-Gilson Argo Climatology	Plain HTTPS from sio-argo.ucsd.edu	Anonymous; small NetCDF files; updated quarterly.
NRL HYCOM ESPC-D v02	THREDDS catalog at tds.hycom.org	Existing HYCOM-on-UAF pipeline could be extended.
NCEP RTOFS 3D	AWS S3 bucket noaa-nws-rtofs-pds + NOMADS THREDDS	Already-mirrored data; only the STAC indexing is missing.
JMA TCHP	JMA NEAR-GOOS HTTP	Lower-priority; coordinate with JMA on access if pursued.

Methodological notes for the AQUAVIEW catalog team

Two systemic findings surfaced during the verification probes:

- 1. **q-search misses items whose titles do not carry common acronyms.** AOML AXBT data are filed as "Expandable Bathythermograph"; q="AXBT" inside NOAA_AOML_HDB returns 0. Adding common acronyms to aquaview:keywords would resolve this.
- 2. **The aggregate endpoint's multi-word q parameter returns 0 even when search_datasets returns thousands of matches.** Aggregate q="ocean heat content" returns 0; search_datasets q="ocean heat content" returns 247,246. Single-word q on aggregate works fine. Recommend either fixing tokenization or documenting the restriction.